

CENG381 Stochastic Processes

Finite Markov Chains - Answer Key 1

Q1 Solution

a) Each row must sum to 1.

$$\text{Row 0: } ? = 1 - 0.4 - 0.2 = 0.4$$

$$\text{Row 1: } ? = 1 - 0.3 - 0.3 = 0.4$$

$$\text{Row 2: } ? = 1 - 0.1 - 0.3 = 0.6$$

$$\begin{array}{c} \\ \\ 0 \\ 1 \\ 2 \end{array} \begin{array}{c} 0 \\ 1 \\ 2 \end{array}$$

b) $P^{(2)} = P * P$ ($p_{ij}^{(2)} = \sum_k p_{ik} * p_{kj}$)

$$p_{00}^{(2)} = 0.4*0.4 + 0.4*0.3 + 0.2*0.1 = 0.16+0.12+0.02 = 0.30$$

$$p_{01}^{(2)} = 0.4*0.4 + 0.4*0.4 + 0.2*0.3 = 0.16+0.16+0.06 = 0.38$$

$$p_{02}^{(2)} = 0.4*0.2 + 0.4*0.3 + 0.2*0.6 = 0.08+0.12+0.12 = 0.32$$

$$p_{10}^{(2)} = 0.3*0.4 + 0.4*0.3 + 0.3*0.1 = 0.12+0.12+0.03 = 0.27$$

$$p_{11}^{(2)} = 0.3*0.4 + 0.4*0.4 + 0.3*0.3 = 0.12+0.16+0.09 = 0.37$$

$$p_{12}^{(2)} = 0.3*0.2 + 0.4*0.3 + 0.3*0.6 = 0.06+0.12+0.18 = 0.36$$

$$p_{20}^{(2)} = 0.1*0.4 + 0.3*0.3 + 0.6*0.1 = 0.04+0.09+0.06 = 0.19$$

$$p_{21}^{(2)} = 0.1*0.4 + 0.3*0.4 + 0.6*0.3 = 0.04+0.12+0.18 = 0.34$$

$$p_{22}^{(2)} = 0.1*0.2 + 0.3*0.3 + 0.6*0.6 = 0.02+0.09+0.36 = 0.47$$

$$\begin{array}{c} P^{(2)} = \\ [] \\ [] \\ [] \end{array}$$

c) $p_{02}^{(3)} = \sum_k p_{0k}^{(2)} * p_{k2}$

$$= 0.30*0.2 + 0.38*0.3 + 0.32*0.6$$

$$= 0.060 + 0.114 + 0.192$$

$$= 0.366$$

Q2 Solution

a) Transition matrix (E=east, W=west):

$$\begin{array}{c} \\ \\ E \\ W \end{array} \begin{array}{c} E \\ W \end{array}$$

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b) $p_{\{EE\}}^{(2)}$ = probability of returning to E after 2 steps starting from E:

$$\begin{aligned} p_{\{EE\}}^{(2)} &= P[E][E]*P[E][E] + P[E][W]*P[W][E] \\ &= (1-p)^2 + p*q \end{aligned}$$

c) Full $P^{(2)}$:

$$\begin{aligned} p_{\{EE\}}^{(2)} &= (1-p)^2 + pq \\ p_{\{EW\}}^{(2)} &= (1-p)*p + p*(1-q) = p(2-p-q) = p(1 - (p+q-1)) \\ p_{\{WE\}}^{(2)} &= q*(1-p) + (1-q)*q = q(2-p-q) \\ p_{\{WW\}}^{(2)} &= q*p + (1-q)^2 \end{aligned}$$

In general $P^{(n)}$ converges to the stationary distribution as $n \rightarrow \infty$, meaning all rows approach the same limiting vector $[\pi_E, \pi_W]$. This suggests that regardless of the starting state, long-run probabilities settle to the stationary distribution.

Q3 Solution

a) Transition matrix P (order S, N, D):

	S	N	D
S	[0.7	0.2	0.1]
N	[0.3	0.5	0.2]
D	[0.1	0.4	0.5]

b) Starting from N: $\mu_0 = (0, 1, 0)$

$$\mu_1 = (0, 1, 0) * P = (0.3, 0.5, 0.2)$$

$$\mu_2 = (0.3, 0.5, 0.2) * P:$$

$$\mu_2[S] = 0.3*0.7 + 0.5*0.3 + 0.2*0.1 = 0.21+0.15+0.02 = 0.38$$

$$\mu_2[N] = 0.3*0.2 + 0.5*0.5 + 0.2*0.4 = 0.06+0.25+0.08 = 0.39$$

$$\mu_2[D] = 0.3*0.1 + 0.5*0.2 + 0.2*0.5 = 0.03+0.10+0.10 = 0.23$$

Distribution after 2 weeks: (S=0.38, N=0.39, D=0.23)

c) $p_{\{SD\}}^{(3)}$ using C-K with $n=2, m=1$:

$$p_{\{SD\}}^{(3)} = \sum_k p_{\{Sk\}}^{(2)} * P_{\{kD\}}$$

First compute row S of $P^{(2)}$:

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$$p_{\{SS\}}^{(2)} = 0.7 \cdot 0.7 + 0.2 \cdot 0.3 + 0.1 \cdot 0.1 = 0.49 + 0.06 + 0.01 = 0.56$$

$$p_{\{SN\}}^{(2)} = 0.7 \cdot 0.2 + 0.2 \cdot 0.5 + 0.1 \cdot 0.4 = 0.14 + 0.10 + 0.04 = 0.28$$

$$p_{\{SD\}}^{(2)} = 0.7 \cdot 0.1 + 0.2 \cdot 0.2 + 0.1 \cdot 0.5 = 0.07 + 0.04 + 0.05 = 0.16$$

$$\begin{aligned} p_{\{SD\}}^{(3)} &= p_{\{SS\}}^{(2)} \cdot P[S][D] + p_{\{SN\}}^{(2)} \cdot P[N][D] + p_{\{SD\}}^{(2)} \cdot P[D][D] \\ &= 0.56 \cdot 0.1 + 0.28 \cdot 0.2 + 0.16 \cdot 0.5 \\ &= 0.056 + 0.056 + 0.080 \\ &= 0.192 \end{aligned}$$